Blessed G/Michael Catholic Secondary School

1st Semester Biology Test One for Grade 12th

Oct, 2018 E.C

Time Allowed: 1 hr

Hanna Ayalew

I. Choose the best answer among the given alternatives

_____1. Which of the following is a method of enhancing food production?
A. animal husbandry C. Tissue culture technologies.
B. Plant breeding D. all

_____2. Which bacteria is the primary starter culture for yoghurt production?
A. Lactobacillus plantarum
B. Streptococcus thermophiles and Lactobacillus bulgaricus
C. Acetobacter acetic
D. Lactobacillus plantarum

_____3. What change in the final fermented milk product is a direct result of lactic acid fermentation?
A. an increase in PH C. a tangy acidic profile
B. increased sweetness D. reduced protein content

_____4. Which of the following is a fermented meat product?
A .Salami B summer sausages C . Country ham D. All

_____5. Which process is characterized by the conversion of sugar or starches into alcohol or acids by microorganisms under anaerobic conditions?
A. fermentation B. pasteurization C. preservation D .freezing

_____6. Southern blotting is ?
A. Attachment of probes to DNA fragments
B. Transfer of DNA fragments from electrophoresis gel to a nitrocellulose membrane
C. Transfer of DNA from nitrocellulose membrane to agarose gel
D. Comparison of DNA fragments from two different sources

_____7. The golden rice variety is rich in?
A Vitamin C B . β-carotene & ferritin C .biotin D . Lysine

_____8, Excision and insertion of a gene is called ?
A. Biotechnology B .Genetic engineering C .cytogenetics D .gene therapy

_____9. Which bacteria is used in the production of insulin by genetic engineering?
A . Saccharomyces B .rhizobium C .E .coli , D .mycobacterium

_____10. The technique used in animal biotechnology for the rapid multiplication and production of animals with desirable traits is ?
A. Hybrid selection and embryo transfer
B. Invitro fertilization and embryo transfer
C. protoplast fusion and embryo transfer.
D .all

_____11. Which one of the following is the third procedure during DNA profiling ?
A . Electrophoresis B .southern blotting
C .restriction endonuclease digestion D. .PCR amplification

_____12. . Mini satellites are ?
A. short repetitive regions on the eukaryotic genome
B. Short non –coding repetitive sequences present throughout the genome
C. short coding repetitive sequences present throughout the genome
D. none

13. In open system during fermentation ?
A. Nutrients are continuously added into the reactor and the products are siphoned off during the run
B .Substrates are added to the system all at once and runs until product is harvested
C . Substrates are added when depleted and the products are finally harvested
D none

14. Which one of the following bacterial species is known for its high rate of biomass production (pruteen)?
A .Xanthomonas B .Methylophilus methylotrophus C .clostridium D all

15. . The process of growing small pieces of plants (explants) in a nutrient medium under sterilized conditions to the whole plant is called ?
A Pluripotency B .totipotency C .multipotency D unipotency

16. . Which technique is used in biotechnology to amplify small segments of DNA ?
A Invivo cloning B .Gel electrophoresis C polymerization chain reaction (PCR) D .southern blotting

17. . Which one of the following products of biotechnology can be produced by the techniques of genetic engineering by using bacteria as a factory to treat heart attack victims ?
A Porcine somatotropin B tissue plasminogen activator C .GH D insulin

18. What is the main goal of gene therapy ?
A To produce medicine in bacteria C .To repair or replace faulty genes
B. to alter the organisms appearance D. to introduce mutations

19. What is bioremediation ?
A. The process of creating genetically modified organisms
B. The process of using bacteria to degrade toxic substances
C .The process of cloning animals
D .The process of tissue culture engineering

20. Which one of the following distilled alcoholic liquor uses grape juice as a source of sugar or substrate for yeast ?
A .bourbon B rum C vodka D .brandy

21. .The process of repeatedly pressing and stretching the dough to increase its elasticity is
A .physical leavening B .biological leavening C .kneading D . Mixing

22. What does a high biochemical oxygen demand indicate in waste water ?
A .low levels of biodegradable organic pollutants
B a high level of dissolved oxygen on the water
C. a large amount of oxidizable organic material
C. The presence of toxic industrial chemicals

23. . What are activated sludge?
A .a mixture of inert materials from primary settling
B .microbial flocs developed through aeration that are used to break down organic matter in sewage
C ..the effluent that is discharged after treatment .
D .the solid waste collected from the bottom of the settling tank

24. .Which one of the following is true during enzyme production in fermentation technology?
A if the enzymes are extracellular then the liquid feed stock is separated from the organism and the enzyme is extracted
B .if the enzymes are intracellular the microorganisms have to be filtered from the feed stock and they are crushed and enzymes are extracted with water
C . A&B
D .none

___25.The bacteria that uses alcohol and organic acids as a source of food to produce acetate .carbon dioxide and hydrogen is ?

A .archaens B .eubacteria C .methanogens D .all

___26. Which cloning technique creates copies of embryonic stem cells that are used in regenerative medicine ?

A .gene cloning B .therapeutic cloning C. .Reproductive cloning D .all

___27. Which one of the following applications of biotechnology related to agriculture is true ?

A .glyphosate resistant gene is introduced into the plant by using Ti plasmid as a vector to develop resistance to herbicides

B . Bt gene is introduced into in maize and cotton by the techniques of genetic engineering to produce plants that kills the larvae of Lepidoptera's

C .B.papillae is used primarily against Japanese beetle larvae D .all

___28. Aerobic microbial driven process that converts solid organic wastes into humus is ?

A .biogas production B composting C biofuel production D .all

___29. The microorganism that ferments all five hexoses that is found in cellulose and hemicellulose into alcohol is ?

A .genetically engineered Saccharomyces cervisae that ferments xylose

B .E.coli that expresses zymogen mobilis gene

C .bakery yeast

D A&B

___30. .What is the waste called after physical removal of solids ?

A .effluent B .secondary treated waste water C .primary treated waste water D .none

___31. .Which of the following is the most common bacteria used in bio mining?

A .Thiobacilus thiooxidans B .Thiobacilus ferroxidans C .Thiobacilus organoparus D .all

___32. An organism that receives genetic material from another organism by the techniques of genetic engineering for the purpose of changing its characteristics is called?

A .cisgenic B . Transgenic C cloned D .mutated

___33. .The most likely detrimental effects on the environment of growing an insect resistant crop are ?

A .The evolution of resistance by insect pests

B .a damaging erect on other species of insects

C .the transfer of the added genes to other species of plants D .all

___34. .Which one of the following genera of fungi are used for as bio pesticides to control the Colorado potato beetle and the froghopper in sugar cane plantations ?

A .Beauveria B .bassiana C .Metarhizium anisopilae D .all

___35. .Plasmids are known as cloning vectors for which of the following reasons ?

A .can be multiplied in culture

B .self replication in bacterial cells

C .can be multiplied in laboratories with the help of enzymes

D .replicate freely outside bacterial cells

___36. What is the primary function of Bacillus thuringensis

A .to solubilize phosphorous in soil

B. To produce an insecticide effective against caterpillars

C .to fix atmospheric nitrogen

D .to act as a predator for insects

37. .What are the advantages of using bio pesticides ?
A .Biopesticides are usually less toxic than conventional pesticides
B .Bio pesticides generally affects only the target pest
C .Bio pesticides often are effective in very small quantities and often decompose very quickly
D .all

38. Nucleopolyhedrovirus is a bio insecticide that is effective against .?
A .bacteria B .fungi C .insect larvae D .all

39. .Which of these is an example of a biological product that can be produced using transgenic animals?
A .antibiotics B enzymes and therapeutic proteins C .synthetic pesticides D .plastic polymers

40. . Which of the following is a common use of transgenic mice ?
A . to produce high fiber foods
B .To serve as a model for studying gene function and disease
C .to test the safety of new vaccines
D . B & c

41. .Restriction endonuclease is also called
A .molecular knife B .target site C .cut ends D ligase

42. .The intentional use of pathogenic bacteria or their toxins as a weapon is known as ?
A..Chemical warfare B .biological warfare C .Conventional warfare D .psychological warfare

43. Which one of the following explains how biology takes part in biosensor technology using biological components such as enzymes and antibodies ?
A .glucose monitoring in blood B .biomedical diagnosis C .environmental monitoring D .all

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Name: Hanna Ayalew Sec: 12TH C R.No 24.

1. D	12. B	23. B	34. D
2. B	13. A	24. C	35. B
3. C	14. B	25. B	36. B
4. D	15. B	26. B	37. D
5. A	16. C	27. C D	38. D C
6. B	17. B	28. B	39. B C
7. B	18. C	29. B C D	40. D
8. B	19. B	30. C	41. A
9. C	20. D	31. B D	42. B
10. B	21. C	32. B	43. D
11. D	22. B C	33. B A D	